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Name - Andryan Durai Jebin

FLOW CONTROL AT DATA LINK LAYER USING SLIDING WINDOW PROTOCOL

AIM:

To write a program to implement flow control at the data link layer using the Sliding Window Protocol and simulate the transmission of frames from one node to another.

ALGORITHM:

1. Start the program.
2. Read the window size from the user.
3. Read a text message from the user.
4. Consider each character of the message as a separate frame.
5. Assign a frame number to each character.
6. Create frames containing the frame number and data.
7. Transmit the frames according to the specified window size.
8. Display the transmitted frames and save them in the Sender_Buffer file.
9. The receiver reads the frames and checks their sequence numbers.
10. If the frames are received correctly, the receiver sends an acknowledgement (ACK).
11. If an incorrect frame is received, the receiver sends a negative acknowledgement (NACK).
12. The sender checks the acknowledgement and transmits the next set of frames or retransmits the required frames.
13. Repeat the process until all frames are transmitted successfully.
14. Stop the program.

PROGRAM:

SENDER PROGRAM (sender.py)

```
import time

window_size = int(input("Enter window size: "))
 message = input("Enter the message: ")
```

```
frames = []

for i, ch in enumerate(message):
 frames.append((i, ch))

with open("Sender_Buffer", "w") as f:
 for i in range(0, len(frames), window_size):
 window = frames[i:i + window_size]

print("\nSending frames:")
for frame_no, data in window:
print("Frame No:", frame_no, "Data:", data)
f.write(f",\n")

f.flush()
 time.sleep(1)

print("Acknowledgement received for the window.")

print("\nAll frames transmitted successfully.")
```

RECEIVER PROGRAM (receiver.py)

```
expected_frame = 0
 received_message = ""

with open("Sender_Buffer", "r") as f:
 frames = f.readlines()

for frame in frames:
 frame_no, data = frame.strip().split(",", 1)
 frame_no = int(frame_no)

if frame_no == expected_frame:
 print("Frame received:", frame_no, "Data:", data)
 received_message += data
 expected_frame += 1
 print("ACK:", expected_frame)
 else:
 print("NACK:", expected_frame)

print("\nReceived message:", received_message)
```

PROCEDURE:

1. Save the sender and receiver programs as sender.py and receiver.py in the same folder.
2. Run the sender program using the command python sender.py.
3. Enter the window size and the message to be transmitted.
4. The sender divides the message into frames and stores them in the Sender_Buffer file.
5. Run the receiver program using the command python receiver.py.
6. Observe the received frames and acknowledgement numbers.
7. Verify that the received message matches the original message.

OUTPUT:

Sender:

Enter window size: 3

Enter the message: HELLO

Sending frames:
Frame No: 0 Data: H
Frame No: 1 Data: E
Frame No: 2 Data: L
Acknowledgement received for the window.

Sending frames:
Frame No: 3 Data: L
Frame No: 4 Data: O
Acknowledgement received for the window.

All frames transmitted successfully.

Receiver:

Frame received: 0 Data: H
ACK: 1
Frame received: 1 Data: E
ACK: 2
Frame received: 2 Data: L
ACK: 3
Frame received: 3 Data: L
ACK: 4

Frame received: 4 Data: O
ACK: 5

Received message: HELLO

RESULT:

Thus, the Sliding Window Protocol was implemented to simulate flow control at the data link layer. Frames were transmitted according to the specified window size, and acknowledgements were used to verify the successful reception of frames.